

Blinkit Sales Analysis

SQL + Power BI Business Intelligence Project

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1. Project Overview

Blinkit is one of India's fastest-growing quick-commerce platforms, delivering groceries and everyday essentials to customers across multiple outlet types and city tiers. With a rapidly expanding footprint, the business generates large volumes of transactional data every day — data that, when analyzed correctly, can reveal powerful insights about what customers are buying, which outlets are performing well, and where operational improvements are needed.

This project takes that raw sales data and turns it into something actionable. Using SQL for data preparation and Power BI for visualization, the goal was to build an end-to-end Business Intelligence solution that gives management a clear, honest picture of how the business is performing — without having to dig through spreadsheets or wait for manual reports.

The final deliverable is an interactive Power BI dashboard that covers everything from high-level KPIs to granular outlet and product-level analysis. It's designed to be used by business stakeholders who need fast answers to business questions, not just data professionals.

2. Business Objective

At its core, this project is about making data useful. Raw sales figures sitting in a database don't tell anyone what to do next. The objective here was to extract, clean, and structure the data in a way that surfaces real answers to the kinds of questions leadership actually asks.

Some of the key questions this dashboard was built to answer include:

- How much total revenue has the business generated across all outlets and product lines?

- Which product categories are driving the most sales, and which ones are underperforming?
- Do customers prefer low-fat products over regular ones, and does this preference vary by outlet?
- Which outlet types — supermarkets, grocery stores, etc. — are the strongest performers?
- Does outlet size (small, medium, large) have a meaningful impact on revenue?
- Which city tiers (Tier 1, 2, or 3) are contributing the most to overall sales?
- How do customer ratings differ across product categories and outlet types?

Answering these questions helps the business make smarter decisions around inventory management, outlet expansion planning, regional marketing strategies, and customer experience improvements.

3. Key Performance Indicators (KPIs)

Four primary KPIs were defined for this dashboard. These metrics act as the scoreboard — the first things a manager or executive should be able to see at a glance before diving deeper into the analysis.

3.1 Total Sales

This is the headline number: total revenue generated from all products sold across every Blinkit outlet during the selected time period. It gives an immediate sense of business scale and performance. Any filter applied to the dashboard (by outlet type, location, or fat content) automatically updates this figure, making it easy to compare performance across different segments.

3.2 Average Sales

Average sales represents the mean revenue per transaction. It's a more nuanced metric than total sales alone — a high total sales figure could simply reflect high volume, while average sales tells you something about the quality of each transaction and how much customers are typically spending per visit.

3.3 Number of Items

This tracks the total count of individual products sold within the chosen filters. It's particularly useful when paired with total and average sales, because it helps distinguish between scenarios where revenue is being driven by a small number of high-value items versus a large volume of lower-value ones.

3.4 Average Rating

Customer ratings are one of the most honest indicators of product quality and satisfaction. The average rating KPI aggregates feedback across all rated products, giving a general sense of how customers feel about what they're buying. Drops in average rating — especially when filtered by product type or outlet — can be early warning signs worth investigating.

KPI	What It Measures
Total Sales	Overall revenue from all outlets and products
Average Sales	Mean revenue per transaction
Number of Items	Total count of products sold
Average Rating	Mean customer satisfaction score across products

4. Business Requirements & Analysis Areas

Based on conversations with business stakeholders, four core analysis areas were identified as priorities for the dashboard. Each one addresses a specific business question and is backed by the KPIs described above.

4.1 Sales by Fat Content

One of the first questions the team wanted answered was whether customers are shifting towards healthier, low-fat options or continuing to prefer regular products. This analysis breaks down total sales by fat content (Low Fat vs Regular) and compares each category across all four KPIs.

The insight here goes beyond just sales volume. Understanding whether low-fat products also receive better ratings, or whether they tend to have higher average transaction values, can directly inform procurement decisions and product placement strategies.

4.2 Sales by Item Type

Not all product categories perform equally, and knowing which ones generate the most revenue is critical for inventory and marketing planning. This section ranks all item types by total sales and also shows supporting metrics like average sales per item type and customer ratings.

The goal is to identify which categories deserve more shelf space and promotional attention, and which may need to be reconsidered. A product category with high volume but low average ratings, for instance, might indicate a quality issue that's limiting its revenue potential.

4.3 Fat Content by Outlet

This analysis adds a layer of depth to the fat content breakdown by comparing customer preferences across different outlet tiers. The question being asked here is: do customers in Tier 1 outlets behave differently from those in Tier 2 or Tier 3 outlets when it comes to fat content preferences?

The answer matters for targeted stocking decisions. If certain outlet types show a clear preference for low-fat products, those outlets should be carrying more low-fat inventory. Generic, one-size-fits-all stocking policies can leave revenue on the table.

4.4 Sales by Outlet Establishment Year

Older outlets naturally have more established customer bases, but that doesn't always mean they outperform newer ones. This section analyzes how revenue varies based on when an outlet was set up, helping the business understand whether establishment year is a meaningful predictor of performance.

For expansion planning, this data is especially valuable. If newer outlets tend to catch up quickly, that's a good sign for aggressive expansion. If older outlets significantly outperform, it might suggest that location selection and time-to-maturity are factors worth accounting for in future investment decisions.

5. Dashboard Visualizations

The Power BI dashboard was structured to tell a story — starting with the overall picture, then progressively drilling into more specific dimensions. Below is a breakdown of the key visualizations included.

5.1 Sales by Outlet Size

A percentage breakdown showing how Small, Medium, and High-sized outlets each contribute to total revenue. This visualization is important for understanding whether Blinkit's current outlet mix is optimal, or whether there's a case to be made for investing in a particular outlet size category.

In many retail businesses, the assumption is that bigger outlets generate more revenue. This visualization either confirms or challenges that assumption with actual data.

5.2 Sales by Outlet Location (City Tier)

A comparison of revenue performance across Tier 1, Tier 2, and Tier 3 cities. This is particularly relevant given that Blinkit operates across a diverse range of Indian cities with very different consumer demographics and purchasing power.

High-performing tiers can be prioritized for additional investment, while underperforming tiers can be examined for operational or marketing improvements.

5.3 Overall Performance by Outlet Type

This is perhaps the most comprehensive visualization in the dashboard — a consolidated comparison of all four KPIs across different outlet types (such as Supermarket Type 1, Supermarket Type 2, Grocery Stores, etc.).

Having all metrics side by side for each outlet type makes it easy to spot patterns. For example, one outlet type might lead on total sales but lag on average rating, indicating a volume-driven model with potential quality trade-offs. Another might show the reverse.

6. Dashboard Features

The dashboard was built with usability in mind. The intention was that someone from the business team — not necessarily a data person — should be able to open it and immediately start getting value from it. The following features support that goal:

- Interactive slicers that allow filtering by outlet type, location, fat content, and item category
- Dynamic KPI cards that update instantly based on the selected filters
- Cross-filtering between visuals, so clicking on one chart filters all others on the page
- Drill-down capability on visuals with hierarchical data
- Clean, professional layout with consistent color coding and typography
- SQL-integrated data model ensuring the numbers in the dashboard match the source data exactly

These features reduce the time it takes for business users to find answers, and they eliminate the need to manually slice data in spreadsheets after the fact.

7. Technology Stack

This project used a combination of tools that are standard in professional data analytics workflows. Each tool was chosen for a specific purpose within the pipeline.

Tool / Technology	Purpose in this Project
SQL	Data extraction, cleaning, transformation, and querying
Power BI	Dashboard development and interactive data visualization
Power Query	Data preparation and pre-processing before loading into Power BI
DAX	Calculating KPIs and defining business measures within Power BI
Excel / CSV	Source dataset containing raw Blinkit sales records

The SQL layer was particularly important for ensuring data quality before anything reached the dashboard. Raw datasets often contain duplicates, inconsistent formatting, or missing values that can distort visualizations if not addressed upstream.

8. Business Insights Generated

While the specific numbers will vary depending on the dataset, the dashboard is structured to consistently surface insights across several dimensions. Below are the types of questions it equips stakeholders to answer confidently:

8.1 Product Performance

- Which item categories are generating the most revenue, and which are lagging?
- Are there product types with high sales volume but low customer ratings that warrant quality reviews?
- How does customer preference for low-fat vs regular products translate into revenue differences?

8.2 Outlet Performance

- Which outlet types are the strongest overall performers when all four KPIs are considered together?
- Does outlet age (establishment year) have a measurable impact on sales performance?
- Are certain outlet sizes consistently outperforming others across locations?

8.3 Geographic Performance

- Which city tiers are contributing most to total revenue?
- Are there regions where sales are underperforming relative to the number of outlets present?
- Do customer purchasing behaviors (fat content preferences, average spend) differ significantly by city tier?

8.4 Operational Opportunities

- Where should the business focus expansion efforts based on existing performance data?
- Which outlets or regions are showing below-average ratings that may need quality intervention?
- What does the data suggest about optimal outlet size and type for new market entries?

9. Conclusion

This project is a practical demonstration of how SQL and Power BI work together to convert raw business data into something genuinely useful for decision-making. The data pipeline — from source CSV through SQL transformation to a live Power BI dashboard — reflects the kind of end-to-end thinking that separates a good analyst from someone who simply makes charts.

The dashboard doesn't just display data. It answers questions. And more importantly, it's built in a way that lets the business ask new questions as they come up, without needing to rebuild anything from scratch. The interactive filtering and cross-visual behavior mean that a single dashboard can support dozens of different analytical conversations.

From a technical standpoint, this project covers a solid range of skills: SQL querying and data preparation, Power Query transformations, DAX measure creation, and Power BI dashboard design. But the real value isn't in the tools — it's in the ability to translate business requirements into an analytical solution that people actually want to use.

The Blinkit Sales Analysis dashboard is a working example of that — and a foundation that can be extended as the business grows and its data needs evolve.